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A MIDDLE/LATE SILURIAN PALEOMAGNETIC POLE FROM LIMESTONE REEFS OF THE WABASH FORMATION, INDIANA, U.S.A

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Abstract. A Middle/Late Silurian paleomagnetic pole has been obtained from reefal limestones in north central Indiana. Eleven of 25 collected sites yielded well clustered characteristic directions after stepwise alternating field demagnetization. One additional site gave the same characteristic direction after great circle analysis. These 12 site-level directions were used to calculate the mean paleomagnetic pole. Reef flank beds dip steeply (about 38°) away from reef centers. Field and petrographic studies demonstrate that part of this dip is a primary depositional dip and that part is due to a tilt caused by early postdepositional differential compaction of underlying mudstone beds. Present attitudes of geopetal surfaces indicate that the postdepositional component of the observed bedding dip is about 12°. A 12° tilt correction to paleomagnetic directions from oppositely dipping flank beds brings the mean directions into significantly better agreement, indicating that the magnetization was acquired early. The presence of both normal and reversed polarities also suggests early acquisition of the characteristic remanence. The mean of the site paleomagnetic poles (with sites from steeply inclined flank beds corrected for 12° of tilt) is 16.8°N, 124.9°E (K=74, A_{95}=5.1°), in good agreement with the Middle Silurian Rose Hill pole for North America.

Introduction

Although the general location of the Paleozoic paleomagnetic apparent polar wander (APW) path for the North American craton has been known since the mid-1950's, many uncertainties still exist with regard to its exact location, and these become increasingly apparent as continental paleomagnetism is called upon to shed light on ever more detailed geological problems. Well constrained Silurian paleomagnetic results for North America are particularly scarce. In a recent review, Van der Voo [1981] lists only three poles that meet generally accepted reliability criteria. These poles are from (1) the Castanea Formation [McElhinny and Opdyke, 1973]; (2) the Bloomsburg Formation [Roy et al., 1967]; and (3) the Rose Hill Formation [French and Van der Voo, 1979]. Of these three, two are from studies more than a decade old, and all are from red beds. In light of recent advances in methodology and instrumentation in paleomagnetism and of the as yet unresolved possibility that many red beds may be remagnetized [e.g., Walker et al., 1981; Larson et al., 1982], additional Silurian results are needed if paleomagnetism is to continue to contribute to our understanding of pre-Mesozoic global geology. The object of the present study is to add to the North American data base with a new Middle/Late Silurian pole from midcontinent limestones.

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Geology and Sampling

Four localities in Silurian reefs were sampled for the present study. However, two of these localities, the quarry near Delphi, Indiana, and the Thornton Quarry near Chicago did not yield meaningful paleomagnetic results. After stepwise demagnetization in the laboratory, directions were either randomly dispersed or clustered about the present field direction. The results from these localities are not considered further except to note that both of these quarries are in dolomite. The other two localities are in limestone.

The two limestone localities are in north central Indiana. One is a small outcrop in a railroad cut near the town of Wabash, and the other is the Pipe Creek Jr. Quarry near Gas City (Figure 1). Both localities are in the Silurian Wabash Formation. The precise age and correlation of the Wabash Formation are currently under study; its age is either Middle or Late Silurian [Shaver and Sunderman, 1983]. Both localities also include exposures of steeply dipping reef flank facies. During deposition of these reef flank units, coarse bioclastic material was shed basinward from the reef core (Figure 2). While this mode of accumulation is relatively straightforward, one additional process has been important: Present dip angles of flank beds are about 38° but range up to 45°, values that are in excess of angles of repose for coarse flank sediment. Several sedimentologists have attempted to explain these anomalously steep dips. Proposed explanations generally fall into two camps. Lehmann [1978] and McGovney [1978], for example, maintain that the present attitude of flank beds is a primary depositional feature developed during the syndepositional stabilization of steep slopes by organism holdfasts and by marine cementation (Figure 3a). Devaney et al. [1983], on the other hand, suggest that while some of the present dip is certainly primary, postdepositional tilting of reef flank beds during differential compaction of underlying mudstones has steepened initial dips to their present values (Figure 3b). Field studies at Pipe Creek Jr. lend strong support to this latter model [Devaney et el., 1983]. Tension fractures, now filled with marine cement and shell debris, developed in response to this differential compaction. The stratigraphically lowest portions of these fractures coincide with the axes of rotation about which postdepositional tilting took place (Figure 3b). In addi-

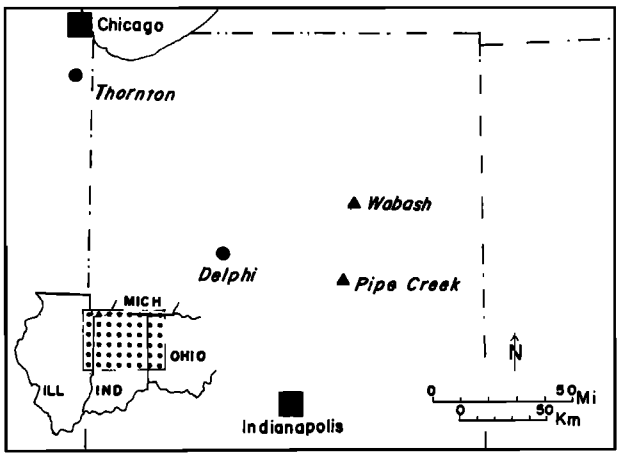


Fig. 1. Map of the study area showing the locations of sampling localities. Triangles are the two limestone localities that yielded meaningful results. [after Shaver and Sunderman 1983].

tion, because fractures are subparallel to the strikes of inclined flank beds, strikes of the flank beds also parallel the axes of rotation.

The orientations of geopetal surfaces (initially subhorizontal surfaces developed during the partial filling of vugs with carbonate mud) were measured in uppermost reef flank facies at Pipe Creek Jr. in order to determine the degree to which postdepositional rotation has steepened flank beds to their present position. In order to measure geopetal inclinations, 12 oriented blocks of limestone were collected from the top of the reef flank sequence (Figure 2) and slabbed so that cuts were vertical and exactly normal to the strike of the beds. As a result, geopetals were exposed on faces normal to axes of rotation.

The significance of this postdepositional rotation of flank beds to the present study is that examination of paleomagnetic signatures of flank units dipping in different directions affords an opportunity to bracket the age of acquisition of magnetization with a "fold" or tilt test [Graham, 1949]. We will return to this test in a later section.

We sampled five stratigraphically defined sites at the Wabash locality and 20 sites at Pipe Creek Jr. for paleomagnetic study. Eight or more oriented samples were taken at each site using a drill and Brunton compass. The sites are in (1) west dipping reef flanks at Wabash, (2) east dipping reef flanks near the top of the quarry at Pipe Creek Jr., and (3) flat-lying mudstones that are exposed at the bottom of the Pipe Creek Jr. Quarry.

Results

Oriented core samples were cut into standard 2.2-cm specimens and measured with an ScT cryogenic magnetometer. Usually one specimen was cut from a sample core, but in some cases, two specimens were obtained. Thermal and alternating field (Af) treatments were attempted on representative specimens; thermal demagnetization results were unacceptably noisy, but Af treatments produced good results in many cases. Therefore the rest of the specimens were Af demagnetized in increments of 5 or 10 mT up to 100 mT or until the magnetization fell below the noise level of the magnetometer (approximately 5 x 10^{-6} A/m). Demagnetization data for each specimen were plotted on orthogonal demagnetization diagrams [Zijderveld, 1967], and the directions of the characteristic magnetizations were determined by vector subtraction. The criterion for isolation of component magnetizations was colinearity of three or more consecutive data points as determined by visual inspection, with subjective allowance for noise.

From representative demagnetization diagrams (Figure 4) it can be seen that directional stability of the characteristic magnetization is only fair in many cases. This instability is not entirely unexpected, however, since intensities of natural remanent magnetization (NRM) before treatment are low, of the order of 10^{-4} A/m, and often drop considerably after treatment at 10 or 20 mT. The rather noisy behavior during demagnetization results in some angular uncertainty in the characteristic direction of a given specimen. This adversely affects the precision parameter k at the site level. However, since deviations due to random noise are themselves random, the mean characteristic direction for the formation (the mean of the site means) should not be affected significantly.

Not all specimens yielded interpretable characteristic magnetizations. When the criterion for determining the direction of the characteristic magnetization (approximate colinearity of

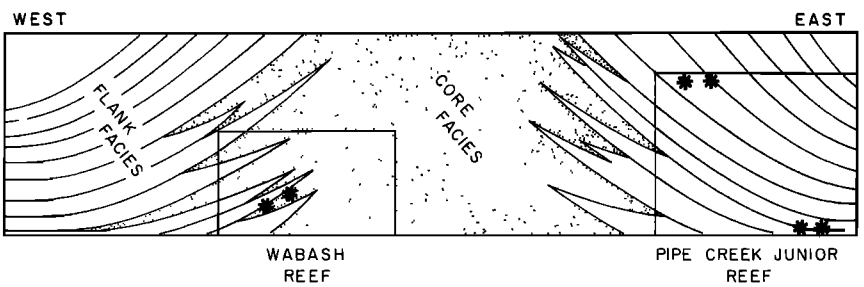


Fig. 2. Cross section through a hypothetical Silurian reef showing component facies and the attitudes of units sampled. Wabash locality samples were taken from west dipping reef flank facies; Pipe Creek Jr. samples were taken from east dipping reef flanks. Delineated areas correspond to areas of core and flank facies at the two exposures; stars designate the units sampled.

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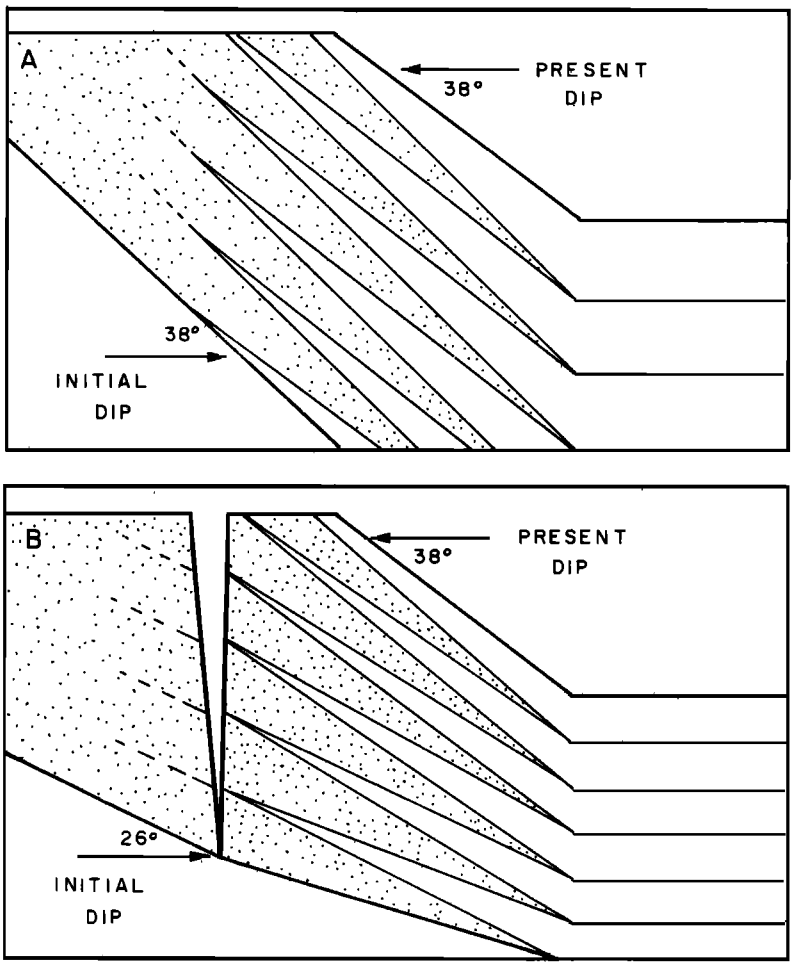


Fig. 3. (a) Schematic model for stabilization of overly steep reef flanks by synsedimentary processes. (b) Alternative model (favored by the authors) showing post depositional development of steepened reef flanks.

three or more data points) could not be met, a specimen was considered to be unstable, and the specimen data were rejected from further consideration. We also found that selection criteria had to be established for data at the site level. Site data were accepted only when the following conditions were met: (1) within-site clustering was significant ($k>3$), (2) the number of accepted specimen directions within a site was four or more, and (3) the site mean direction was significantly different from the present field direction. After these criteria were applied, 11 (of 25) site results were found acceptable.

In addition, one site that gave dispersed directions yielded a nonpresent field direction after analysis of remagnetization great circles [Halls, 1976]. This site-level result was included in the calculation of the final mean paleomagnetic pole.

In order to apply a tilt test to bracket the age of the magnetization it was necessary to obtain an estimate of the amount of postdepositional tilting that has steepened inclined reef flank beds to their present attitude. Orientations of 1493 geopetal surfaces were measured for this purpose. These data were obtained from three geopetal fabrics: (1) fibrous cement/micrite contacts in large stromatactic voids (534 geopetals), (2) spar cement/micrite contacts in shelter voids within and immediately below large shell fragments (481 geopetals), and (3) spar/micrite contacts in fenestral voids (478 geopetals). Significantly, all three types of fabrics must record early void filling soon after deposition. Because fibrous cement in reefal sequences records carbonate precipitation from marine fluids [e.g., Bathurst, 1975], its presence in the first fabric type requires void filling during flank bed deposition. Because shelter cavities and fenestral voids are both primary pore types in such sequences, partial filling of these voids by marine micrite must also have been early.

Geopetals from all three void types have similar orientations, ranging from contacts that dip 20° toward the reef core to those that dip 50° toward the basin. In addition, all three geopetal populations exhibit Gaussian distributions with means inclined about 12° toward the basin. Figure 5 is a histogram showing the distribution of all geopetal orientation data. While this 70° range in inclinations seems large, it is nearly identical to the only other large geopetal data set reported to date. These are data from similar facies and void types in the Devonian

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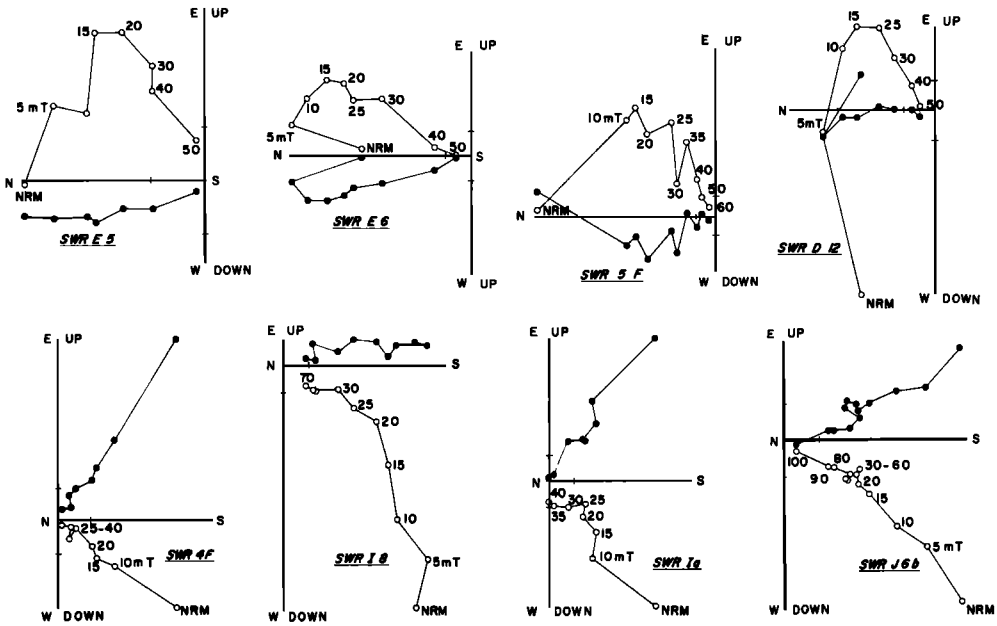


Fig. 4. Representative demagnetization diagrams [Zijderveld, 1967]. Open circles are projections of the end points of the magnetization vectors onto a north-south vertical plane; solid circles are the projections onto the horizontal plane. The tic marks on the axes are 10^{-5} A/m. (Note that the noise level of the magnetometer is about one half of a tic mark.) (Top) Normally magnetized specimens. (Bottom) Reversely magnetized specimens.

barrier reef of the Canning basin in Australia [Playford, 1980]; here, geopetal contacts range from those that dip 22° toward the reef core to those that dip 48° toward the basin, with a mean inclination of 6.3° toward the basin [Playford, 1980]. Because data from the Devonian of Australia and the Silurian of Indiana, reported here, are the only large data sets on geopetal orientations of which we are aware, we believe them to be representative of geopetal orientations in reefal facies. These data sets indicate that apparently even one generation of void-filling micrite may result in a range of geopetal orientations, probably because the sediment that ultimately fills these voids may lap onto void walls. Although the mean orientation of such mud surfaces should reflect the paleohorizontal, much variation exists. Because (on average) these horizons now dip basinward at 12° (Figure 5), that much tilt must have taken place since the upper flank beds of the Pipe Creek Jr. Quarry were deposited.

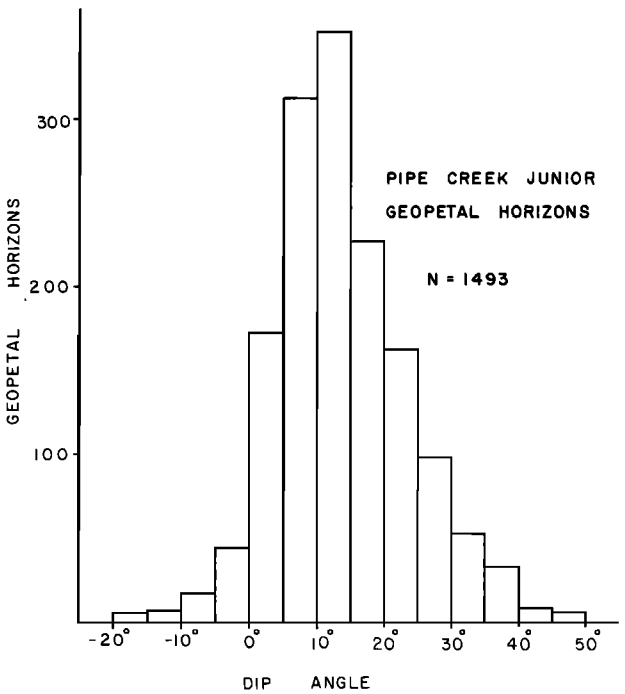


Fig. 5. Histogram of geopetal dips from the top of the Pipe Creek Jr. Quarry. Positive (negative) dip angles are basinward (reefward). Mode dip is 12° basinward.

For the tilt test we separated specimen characteristic directions from steeply inclined beds into two populations: one from west dipping flank beds at Wabash and the other from east dipping beds at Pipe Creek Jr. (Figure 2). The 12° tilt correction obtained from the average dip of geopetal horizons was used as the best estimate of the postdepositional component of tilt. The horizontal axis about which tilt was rotated at Pipe Creek Jr. was the strike of a major tension fracture trend, a trend which is also subparallel to bedding strike. At Wabash, where no tension fractures are exposed, the bedding strike was used as the rotation axis. Usual fold tests [e.g., McElhinny, 1964; McFadden and Jones, 1981] cannot be applied in this case because the number of observations and the precision parameters for the two populations are significantly different. However, the 95% confidence limits about the means of the Wabash and Pipe Creek populations do not overlap before tilt correction, but they do after correction

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TABLE 1. Results of the Paleomagnetic Tilt Test

Tilt Correction						Direction			
						Before Correction		After Correction	
Strike	Dip	N	k	α_{95}	R	D,deg	I,deg	D,deg	I,deg
180°	12°W	15	42.7	5.9	14.7	138.9	40.3	148.2	47.4
326°	12°E	37	17.7	5.8	35.0	156.6	41.3	145.8	42.4
		52	19.3	4.6	49.4	151.3	41.3		
		52	21.2	4.4	49.6			146.5	43.9

Row 1 contains the means of specimen directions from west dipping flank beds at the Wabash locality, row 2 those from east dipping beds at Pipe Creek Jr., and rows 3 and 4 those from all reef flank directions before and after tilt correction respectively. D is declination, I is inclination, and N is the number of specimen directions used in the analysis. k and α_{95} are statistical parameters and R is the length of the resultant vector [Fisher, 1953].

(Table 1 and Figure 6). It is therefore likely that the magnetizations were acquired early, before compaction of underlying mudstone units. Since tension fractures associated with compaction are lined with marine cement precipitated immediately after deposition and in some cases contain Devonian clastic and shell debris [Lehmann, 1978], the magnetizations were acquired early, perhaps during deposition.

After correction for postdepositional tilt the 93 specimen characteristic directions cluster reasonably well into two nearly antipodal groups, one northwesterly and up, the other southeasterly and down (Figure 7a). Site-level data are listed in Table 2 and plotted in Figure 7b (with normal directions shown reversed). The mean of the site virtual geomagnetic north poles is located at 16.8°N, 124.9°E (K=74; A_{95}=5.1°).

Acquisition of isothermal remanent magnetization (IRM) experiments were carried out on representative specimens. In all cases, IRM was acquired rapidly, and saturation was reached below 300 mT. The low coercivities evident in these experiments and in the Af demagnetization of NRM indicate that magnetite is the important carrier of NRM. The origin of the magnetite (detrital, biogenic, or diagenetic) is not as yet clear. Magnetic extracts have been prepared and analyzed using methods described by McCabe et al. [1983], but the results are inconclusive. Further work is in progress.

Discussion

An ancient magnetization has been isolated in roughly half of the Wabash Formation sites sam-

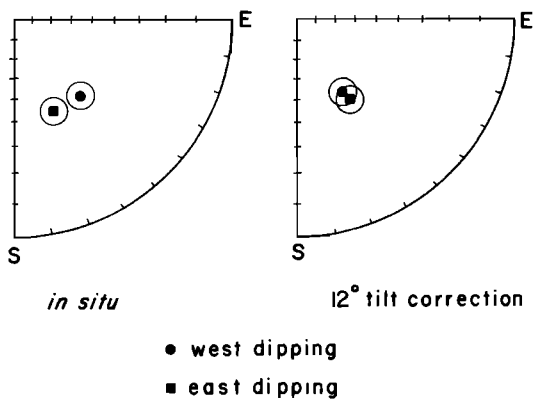


Fig 6. Graphic illustration in equal-angle projection of the tilt test to bracket the age of the magnetization. Mean directions from two populations are shown before and after correction for 12° of tilt. One population is composed of all specimen directions from east dipping flank beds, the other those from west dipping flank beds. Circles about the means are the α_{95}'s. Mean directions are in significantly better agreement after the tilt correction is applied.

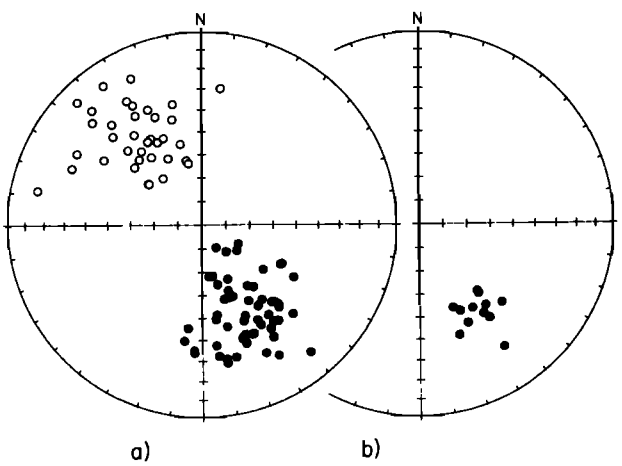


Fig. 7. (a) Specimen characteristic directions in equal-area projection from 11 accepted sites (93 observations) after tilt correction. (b) Site-level directions after tilt correction (with normal directions reversed) for the 11 sites represented in Figure 7a plus the one site analyzed with great circle convergence. Projection as in Figure 7a.

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TABLE 2. Summary of Site-Level Paleomagnetic Results

Site	N/N_0	R/N	k	α_{95}	Direction		Pole	
					D,deg	I,deg	°N	°E
4	15/15	15/15	42.7	5.9	148.2	47.4	14.7	122.7
5	8/8	0/8	24.5	11.4	155.4	48.8	16.2	116.5
6	5/8	2/5	13.2	21.9	144.8	42.6	17.1	127.6
A	6/9	0/6	41.9	10.5	143.6	39.3	18.8	129.9
D	5/8	0/5	9.1	26.8	141.3	45.0	14.0	129.6
E	7/10	1/7	24.6	12.4	140.1	51.2	8.9	127.9
N	6/10	0/6	40.0	10.7	145.9	24.0	28.3	132.8
G2	10/16	8/10	14.5	13.1	139.8	52.4	7.8	127.5
H					134.0	41.0	13.1	137.0*
I	12/18	11/12	19.3	10.2	158.7	51.5	14.8	112.9
J	10/14	9/10	15.8	12.5	154.3	42.4	20.8	119.4
K	9/12	9/9	18.2	12.4	160.8	38.8	25.2	114.1

N is the number of specimen directions used in the analysis, N_0 is the number of specimens demagnetized, D is declination, I is inclination, and R is the number of reversed-polarity characteristic directions. K, k, A_{95}, and α_{95} are statistical parameters [Fisher, 1953]. Site 4 is from west dipping reef flanks at the Wabash locality. Sites 5, 6, A, D, E, and N are from east dipping upper reef flanks at Pipe Creek Jr. Sites G2, H, I, J, and K are from flat-lying lower mudstones at Pipe Creek Jr. Directions from flank beds corrected for 12° of tilt (see text). Specimen directions given unit weight. The mean paleomagnetic pole is located at 16.8°N, 124.9°E (N=12, K=73.7, A_{95}=5.1°).

* Direction from remagnetization circle analysis.

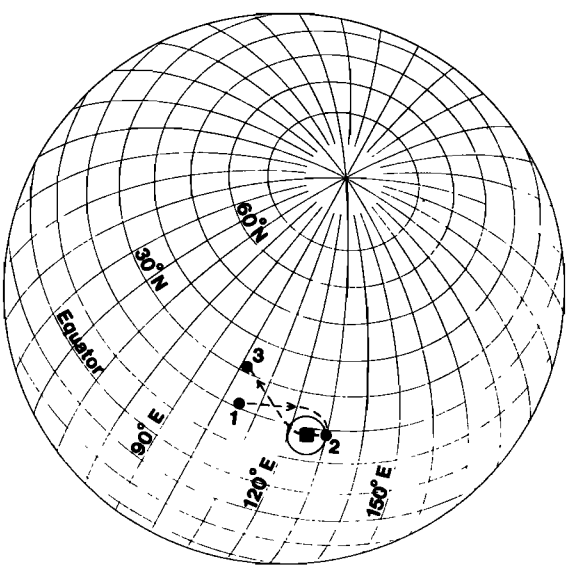


Fig. 8. Silurian paleomagnetic poles from cratonic North America. Dashed curve connects poles in their stratigraphic order. Numbered poles are as follows: (1) Castanea (Early Silurian); (2) Rose Hill (Middle Silurian); (3) Bloomsburg (Late Silurian). Square is the Wabash Formation pole (Middle/Late Silurian, this study) surrounded by its 95% cone of confidence.

pled, and a tilt test indicates that the magnetization was acquired early. The presence of both normal and reversed characteristic directions (Table 2 and Figure 7a) is additional (albeit inconclusive) evidence for an early acquisition of magnetization.

The paleomagnetic pole obtained in this study is plotted in Figure 8 along with other Silurian poles for North America. The dashed curve in Figure 8 is the Silurian APW path if all these poles are the result of successfully isolated single component magnetizations that were acquired in stratigraphic order. Alternative (and perhaps simpler) APW paths may be more valid if any of these poles (or their ages) can be shown to be faulty. While it is not possible to resolve this problem based on presently available information, we note that the pole from the present study falls very close to the Rose Hill pole [French and Van der Voo, 1979], and tends to confirm that result.

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